# DEVELOPMENT OF A WEB-BASED STUDENT GROUP FORMATION AND SKILL MATCHING SYSTEM

**BSIT Class Project Concept Paper**

**John Carl V. Cuayzon | Ryan M. Balinas | John Cris G. Egca**
Bachelor of Science in Information Technology (BSIT 3A)

## INTRODUCTION

### Background of the Project

Collaborative learning and group-based capstone projects are fundamental components of the Bachelor of Science in Information Technology (BSIT) curriculum. In typical academic settings, instructors assign students to project teams through arbitrary methods such as randomized grouping, student self-selection, or static roster segmentation. However, these traditional grouping mechanisms consistently present major operational challenges. Student self-selection often leads to groups formed strictly along social affinities rather than balanced technical proficiencies, leaving less socially connected students isolated. Furthermore, arbitrary or random distribution frequently produces severely unbalanced teams—either concentrating multiple proficient programmers in a single team or forming groups that completely lack critical competencies, such as database design, user interface engineering, or systems analysis. When a project team discovers these competency deficits mid-semester, projects frequently suffer from uneven workloads, delayed milestone submissions, interpersonal conflicts, and compromised deliverable quality.

To address these inefficiencies, the proponents have initiated the design and implementation of a centralized, web-based Student Group Formation and Skill Matching System. Built upon Python and the Django framework, the platform digitizes student technical profiling, catalogs course project requirements, and establishes standardized skill inventory matrices. Building upon the currently implemented database architecture, student profiling modules, and project skill requirement registries, the final system will incorporate algorithmic skill matching, automated group gap analysis, and intelligent member recommendations. This Information Technology solution will eliminate manual guesswork for instructors, foster balanced student collaboration, and ensure that academic project teams possess the complementary competencies necessary to successfully complete complex software systems.

### Statement of the Problem or Project Objectives

The main objective of this project is to design, develop, and evaluate a web-based Student Group Formation and Skill Matching System to automate the identification of student technical proficiencies, match complementary skill sets against project requirements, and optimize collaborative team formation for academic coursework.

Specifically, the project aims to accomplish the following technical objectives:

1. **Maintain and extend a secure student and skills management repository** that captures student academic profiles, technical skills, and standardized proficiency ratings across a five-point evaluation scale *(implemented in the current phase)*.
2. **Maintain and extend a project requirement definition module** that records specific project specifications and maps prerequisite skill levels required for successful project execution *(implemented in the current phase)*.
3. **Develop an algorithmic skill matching engine** that will compute quantitative compatibility indices between student skill inventories and specified project requirements *(planned for final development)*.
4. **Implement an automated group formation and skill gap identification module** that will organize students into balanced teams, analyze group rosters against project requirements, and flag missing proficiencies *(planned for final development)*.
5. **Build an intelligent group member recommendation feature** that will suggest optimal candidate students to fill identified skill deficits within existing groups based on complementary proficiencies *(planned for final development)*.
6. **Establish role-based authorization and session security** (Administrator, Instructor, and Student) with comprehensive automated testing to evaluate system usability, reliability, and algorithmic accuracy.

**Significance of the Project**

The proposed system will provide direct practical value to the following beneficiaries:

- **Course Instructors and Capstone Advisers:** Streamlines the labor-intensive administrative process of group formation, eliminates subjective bias in team assignments, provides transparent visibility into class-wide technical proficiencies, and highlights struggling groups before project deadlines.
- **BSIT Students:** Promotes equitable project collaboration by ensuring fair distribution of technical skills across all groups, minimizes interpersonal friction caused by mismatched expectations, and helps students identify their individual technical strengths and growth areas.
- **Academic Departments and College Administrators:** Provides objective, aggregated metrics on student competencies across diverse technical domains (e.g., frontend development, backend engineering, database management), serving as empirical data for curriculum reviews and instructional interventions.
- **Future Researchers and Student Developers:** Serves as a modular architectural reference for applying heuristics, algorithmic matching logic, and role-based data architectures within higher education web applications.

**Alignment with Institutional Research Agenda and Sustainable Development Goals**

- **Institutional Research Agenda Alignment:** Aligns with the Institutional Priority on *Educational Technology and Academic Information Systems Modernization* *[Note: Exact agenda title to be confirmed with the class adviser]*. The project modernizes collegiate

administrative and instructional processes through automated web-based skill tracking, data-driven academic grouping, and decision-support algorithms.

- **Sustainable Development Goal Alignment:**
  - **SDG 4 (Quality Education — Target 4.4):** Fosters relevant technical and collaborative skills by ensuring balanced peer-learning environments where students develop complementary competencies, directly supporting effective learning outcomes in higher-education computing programs.
  - **SDG 9 (Industry, Innovation, and Infrastructure):** Fosters technological innovation within academic infrastructure by replacing outdated, paper-based, or ad-hoc group selection practices with an algorithmic digital platform.
  - **SDG 10 (Reduced Inequalities):** Promotes equitable opportunity and reduces social bias in academic project formation by utilizing objective skill proficiencies rather than social cliques, preventing marginalization of individual learners.

## SCOPE AND LIMITATIONS

### System Scope & Technical Architecture

The system is designed as an institutional web-based platform tailored for academic departments conducting group-based computing projects. The functional modules and technical architecture are summarized in the table below:

| Component | Description & Coverage |
|---|---|
| **Core Modules** ***(Currently Implemented)*** | • **Authentication & Session Security:** User login with rate limiting and lockout protection, account registration with password validation, logout, and superuser user management.<br>• **Student Records Management:** Searchable, paginated student profile directory with modal-driven AJAX CRUD operations (student number, name, email, course, section, year level 1–4).<br>• **Skills Catalog Management:** Master repository of technical competencies with descriptions and modal-driven AJAX CRUD operations.<br>• **Student-Skill Proficiency Assignment:** Normalized mapping of students to skills with calibrated proficiency levels (1 = Beginner to 5 = Expert) and duplicate prevention.<br>• **Project & Prerequisite Skill Management:** Project listing and assignment of specific required skills with minimum expected proficiency ratings.<br>• **Django Administration & Automated Testing Suite:** Administrative data controls and in-memory test runner validating models, URLs, and view operations. |
| **Core Modules** ***(Planned for Final Phase)*** | • **Group Creation & Roster Management:** Interactive interface for instructors to define project groups, set member quotas, and assign student rosters.<br>• **Algorithmic Skill Matching Engine:** Service algorithm that will calculate compatibility scores between individual student profiles and |

| Component | Description & Coverage |
|---|---|
| | project requirement matrices.<br>• **Skill Gap Analysis Module:** Automated evaluation tool that will compare collective group competencies against project requirements to identify skill shortages.<br>• **Intelligent Member Recommendation Engine:** Heuristic recommender that will suggest candidate students to balance team competencies.<br>• **Analytics & Dashboard Overview:** Visual metrics dashboard presenting student totals, skill distribution charts, project rosters, and group statuses. |
| **Technology Stack** | • **Backend:** Python 3 with Django 5.0 (Model-View-Template architecture, secure ORM, session handling).<br>• **Database:** MySQL / MariaDB for relational persistent storage (`student_group_matching`), with SQLite utilized for in-memory automated test suites.<br>• **Frontend:** HTML5, CSS3, JavaScript (asynchronous Fetch/AJAX APIs for dynamic modal interactions), Bootstrap 5, Tabler UI Dashboard Framework, and Font Awesome 6 icons.<br>• **Testing:** Django Test Framework and Pytest runner with in-memory schema generation. |
| **Target Users** | • **Course Instructors / Advisers:** Formulate projects, specify skill requirements, inspect group compatibility, generate automated groups, and monitor group composition.<br>• **BSIT Students:** Self-report or update verified skill proficiencies, view project requirements, and view assigned project teams.<br>• **System Administrators:** Manage user accounts, role permissions, and system configurations. |

## Limitations

- **Deployment Environment:** The system will operate within an institutional intranet, local campus network, or designated web server environment configured for academic laboratory classes.
- **Scope of Matching Logic:** The matching engine will evaluate structured technical proficiencies (levels 1 through 5) and required project skill profiles. It will not assess subjective psychological personality indicators, psychometric temperament tests, or scheduling availability constraints in the current development phase.
- **Assessment of Student Skills:** Student skill proficiency ratings will rely on student self-declaration and instructor verification; automated external coding assessments, live code execution sandboxes, and third-party credential verification APIs are excluded from the scope.
- **External Communications:** Mobile SMS gateway alerts, email dispatch daemons, and external Learning Management System (LMS) synchronization protocols (e.g., Canvas, Moodle LTI) are excluded from the current development phase and may be recommended for future feature updates.